

Note on A course in Arithmetic by J.-P. Serre

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1 Chapter 1

This is a list of proofs of nontrivial results in Chapter 1 of *A course in Arithmetic* by J.-P. Serre.

Let K be a finite field with q elements.

Theorem 1.1

Let u be an integer ≥ 0 . The sum

$$S(X^u) = \sum_{x \in K} x^u = \begin{cases} -1 & \text{if } u > 0 \text{ and } (q-1) \mid u \\ 0 & \text{otherwise} \end{cases}$$

$u = 0$ is subtle here. We define $x^0 = 1$ for all $x \in K$ (this definition is nice when we work with polynomials because we can write 1 as x^0 to maintain symmetry), so $S(X^0) = q = 0$.

The proof is easy. Use primitive roots.

The variant proof: $x \mapsto x^u$ is a group homomorphism from $K^\times$ to itself. The kernel is all x such that x is a root of $x^{\gcd(u, q-1)} - 1$. So the image has size $\frac{q-1}{\gcd(u, q-1)}$, which implies the image is just the $d = \frac{q-1}{\gcd(u, q-1)}$ roots of unity. So the sum is 0 unless $d = 1$ (sum of d -th roots of unity is 0 if $d \geq 2$ is coprime to p).

Theorem 1.2 (Chevalley-Warning Theorem)

Let $P_1, \dots, P_r$ be polynomials in n variables over a finite field K of characteristic p . If the sum of their degrees is less than n , then the number of common solutions to the equations $P_1 = \dots = P_r = 0$ is divisible by p .

Proof. This uses a common trick in counting roots over finite fields. Consider the polynomial

$$Q = \prod_{i=1}^r (1 - P_i^{q-1})$$

The number of common solutions is equal to $\sum_{x \in K^n} Q(x)$ since if $P_i(x) = 0$ for all i , then $Q(x) = 1$, otherwise $Q(x) = 0$.

Expand this. For each monomial in the expansion $x_1^{u_1} \cdots x_n^{u_n}$, since $\sum \deg(P_i)(q-1) < n(q-1)$, we have u_j is not divisible less than $q-1$ for some j . So by the previous theorem, the sum over K^n of this monomial is 0. (that's why we define $x^0 = 1$ above). So the number of common solutions is 0 mod p . $\square$

Corollary 1.3

All quadratic forms in at least 3 variables over a finite field have nontrivial zeros.

Theorem 1.4

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$$

Proof. Let α be the primitive 8th root of unity α in $\overline{\mathbb{F}_p}$. Then, $y^2 = (\alpha + \alpha^{-1})^2 = 2$. If $p \equiv \pm 1 \pmod{8}$, then $y^p = y$, so 2 is a square in $\mathbb{F}_p$. If $p \equiv \pm 3 \pmod{8}$, then $y^p = -y$, so 2 is not a square in $\mathbb{F}_p$. $\square$

Theorem 1.5 (Law of Quadratic Reciprocity)

For two odd primes p and q , we have

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

The key is to consider the Gauss sum: Let ζ be a primitive p -th root of unity in $\overline{\mathbb{F}_p}$, then define

$$y = \sum_{x \in \mathbb{F}_p} \left(\frac{x}{p}\right) \zeta^x$$

Then we have $y^2 = \left(\frac{-1}{p}\right) p = (-1)^{\frac{p-1}{2}} p$ and $y^{q-1} = \left(\frac{q}{p}\right)$. Combining these two gives the result.

2 Meeting 1 on Jan. 28 2026

This is a list of nontrivial results in the first meeting of the course.

Theorem 2.1

The finite subgroup of a field's multiplicative group is cyclic.

This can be easily proven using the structure theorem of finite abelian groups and the fundamental theorem of Algebra.

Definition 2.2. For any prime p or $p = \infty$, we define the

$$\|x\|_p = \begin{cases} p^{-v_p(x)} & p \text{ is a prime} \\ \|x\| & p = \infty \end{cases}$$

Theorem 2.3

For any rational number r , $\prod_p \|r\|_p = 1$

This is trivial from the definition of v_p .

We can in general define a norm on a number field K as follows:

Definition 2.4. $\|\cdot\| : K \rightarrow \mathbb{R}_{\geq 0}$ is a norm if

1. $\|x\| = 0$ iff $x = 0$
2. $\|xy\| = \|x\| \|y\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Corollary 2.5

$\|1\| = 1$ and $\|-1\| = 1$

Definition 2.6. Norm N_1 and N_2 are equivalent if there exists a real number c such that $N_2(x) = N_1(x)^c$ for all x .

Theorem 2.7 (Ostrowski's Theorem)

Any non-trivial norm on $\mathbb{Q}$ is equivalent to either the usual absolute value or to $\|x\|_p$ for some prime p .

Proof.

□

Theorem 2.8 (Hensel's Lemma)

Proof.

□

Theorem 2.9 (Hasse principle)

Example 2.10

$f(x, y, z) = 5x^2 + 7y^2 - 13z^2$. We want to find a nontrivial $f(x, y, z) = 0$ has non-trivial solutions in $\mathbb{Q}$ using Hensel's Lemma.

Consider this curve in $\mathbb{P}^3$, is there any other solution?

Use parameterization of this curve.

Theorem 2.11

If $n \geq 3$, $x^n + y^n = z^n$ has no non-trivial solutions in $\mathbb{C}[x]$.